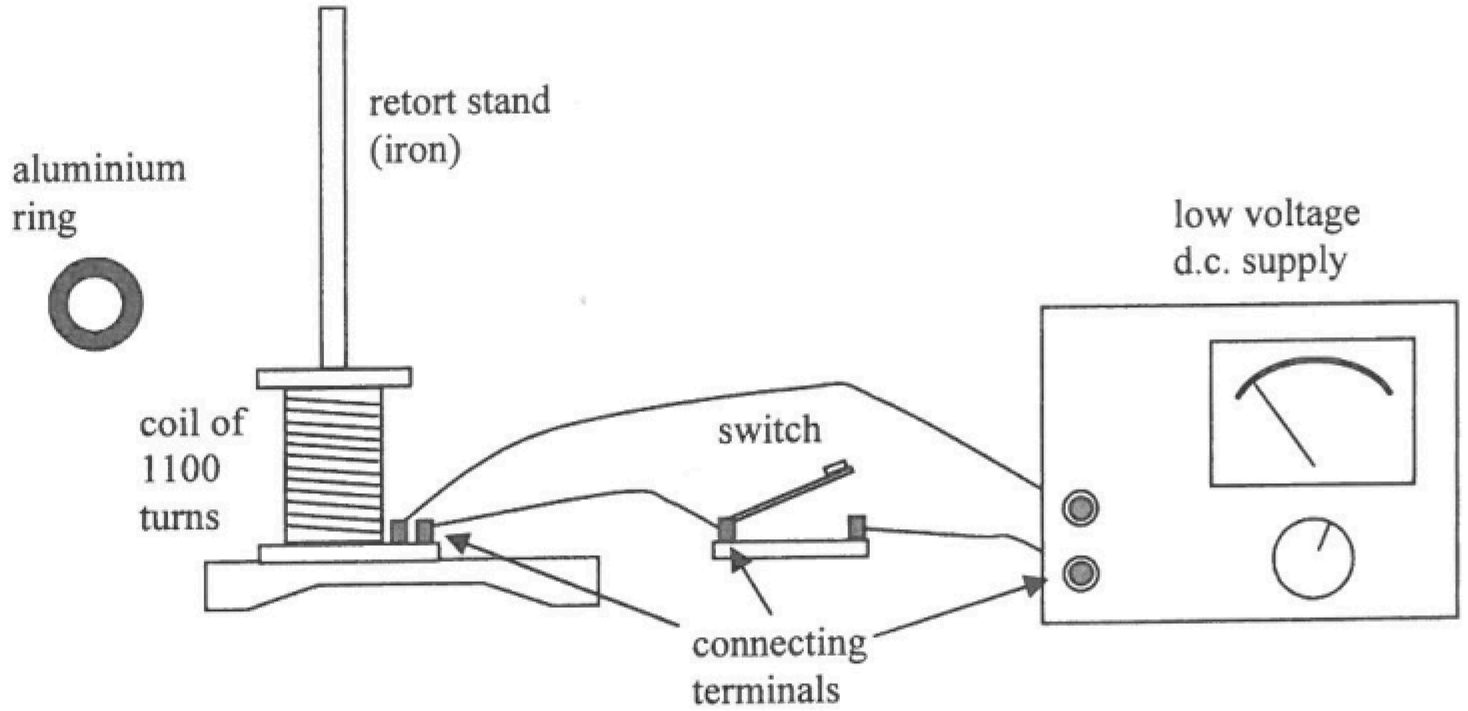
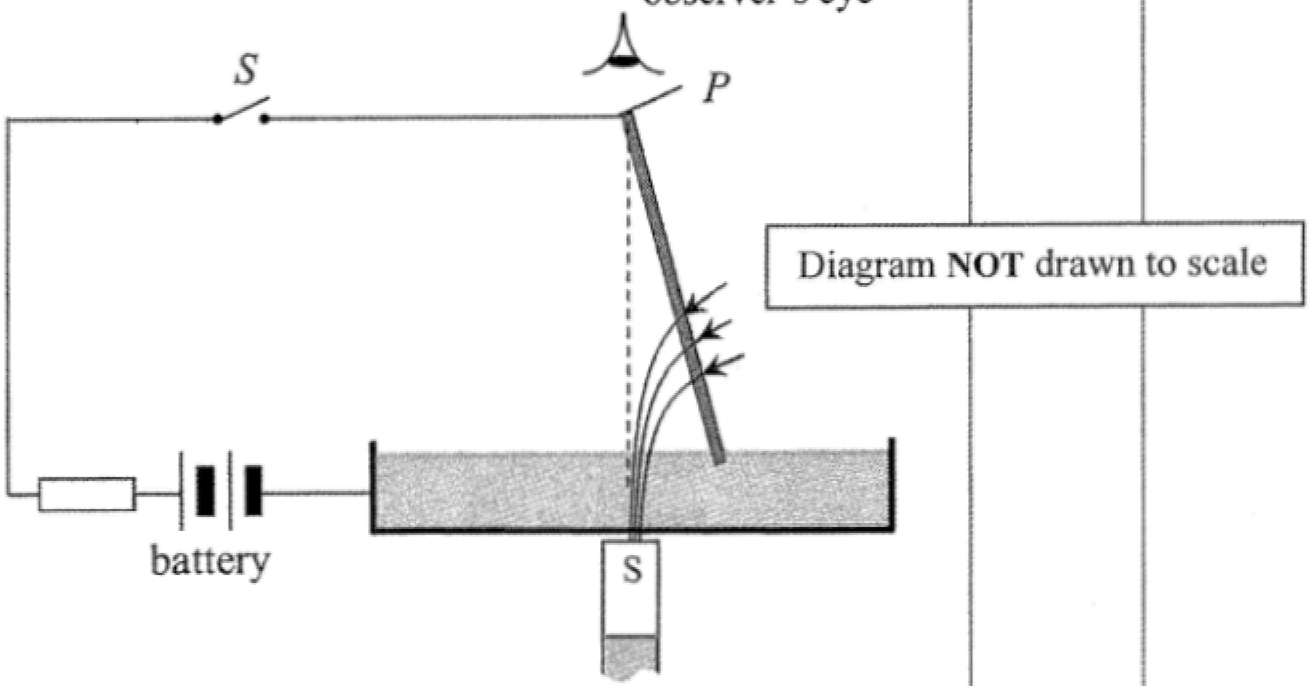
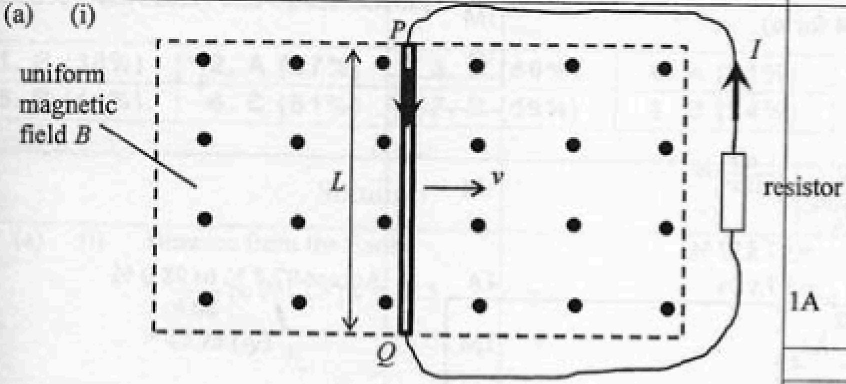
9.	(a)		Humid conditions in bathrooms and water is a conductor which provides a conducting path or lowers the resistance between our hands/body and the source of electricity.	1A 1A	2
	(b)	(i)	The person will get electric shock because full potential drop 220 V is applied to or substantial/large current flows through the human body.	1A 1A	2
		(ii)	The person will not get electric shock / Nothing happens because the current in secondary circuit has no return path/incomplete circuit.	1A 1A	2
	(c)		Primary : Secondary = 2:1 for 110 V	1A	1

9. (a)	Connect the coil to the terminals of the d.c. supply via the switch (diagram).	1A	



Put the aluminum ring on the top of the coil through the rod of the retort stand.	1A	
When closing the switch, the ring would shoot up the rod once,	1A	
as the aluminum ring experiences a changing magnetic field produced by the coil at the start,	1A	
According to Lenz's Law, eddy currents flow in the ring to oppose the change.	1A	
Or A magnetic field in the opposite direction generated by the eddy currents repels the magnetic field produced by the coil.	1A	
However, when the current and thus the resulting magnetic field are constant, the ring would fall back to the coil as eddy currents no longer flow.	1A	
		6
(b) (i) The aluminum ring would float in the air.	1A	
		1
(ii) The aluminum ring with a slit would remain stationary.	1A	
		1

9.	(a)	Right (current flowing downward, B-field into paper) When the rod reaches the highest point it falls. Then its lower end touches the conducting liquid again and the same magnetic force makes it 'kick' out from the liquid. The process repeats so the rod continually 'kicks' out and then returns.	1A 1A 1A	
				3
	(b) (i)	As moment = $F \times d$ 7.2×10^{-4} N m = F (0.09 m) $F = \dfrac{7.2\times10^{-4}}{0.09} = 8.0 \times 10^{-3}$ N	 1M 1A	
				2
	(ii)	$F = BIl$ 8.0×10^{-3} N = B (3.2 A) (0.06 m) $B = 0.042$ T	1M 1A	
				2
	(c) (i)	Correct sketch.	1A	
				
				1
	(ii)	The rod will rotate anti-clockwise (as viewed from above).	1A	
		Or The rod describes a conical pendulum.	1A	
				1

Solution	Marks	Remarks
9. (a) (i) 	1A	
	1	
(ii) By Lenz's law, a magnetic force F_B acts on the rod such that it opposes the rod's motion,	1A	Accept : Work done by an external force is needed to transfer mechanical energy into electrical energy.
an external force F is needed to balance F_B so as to maintain uniform motion (or constant v)	1A	
$\therefore F = F_B = ILB$ (in magnitude)	1A	
Or $F \cdot v = I\xi$, $\therefore F = \frac{I\xi}{v}$	1A	
	3	
(iii) mechanical power input $= Fv$ $= (ILB)\,v$	1M	
power input = power output (electrical) $ILB\,v = I\,\xi$ $\xi = BLv$	1M	
	2	
(b) (i) Horizontal (component) is perpendicular to the mast.	1A	
Or Vertical (component) is parallel to the mast.	1A	
	1	
(ii) $\xi = (B\cos 30°)\,L\,v$ $= (50 \times 10^{-6}\cos 30°)\,(20)\,(6)$ $= 5.196152 \times 10^{-3}$ V	1M	by (a)(iii) using the horizontal component of B
≈ 5.20 mV	1A	
more electrons at end X	1A	
	3	
(iii) No current.	1A	
Both the cable and the mast cut the field lines in the same way, the e.m.f.'s produced are equal and thus oppose each other.	1A	
Or No change in magnetic flux through the loop of the circuit.	1A	
	2	

7.	(a)	(i)	Eddy currents are induced to oppose the change of magnetic flux due to the movement of the metal sheet. Moving to the left → accept magnetic field	1A 1A	
				2	
		(ii)	Kinetic energy → Electrical energy → Internal (heat/thermal) energy	2A	
				2	
		(iii)	Limitation: Eddy braking only works when the vehicle is moving. e.g. The vehicle cannot park in stationary positions / still on a slope.	1A	
				1	
	(b)		Electrical energy consumed = $2\ \text{kW} \times \frac{15}{60}\ \text{h} = 0.5\ \text{kW h}$ Cost = $\$\ 1.1 \times 0.5 = \$\ 0.55$	1M 1A	
				2	
	(c)		Lamination, i.e. the cores of these devices are made up of multiple insulated sheets of metal.	1A	
				1	
	(d)		The magnetic field (produced by eddy currents) measured will decrease (where there are irregularities). It is because the crack will reduce the eddy currents.	1A 1A	
				2	